CS 578 Statistical Machine Learning (Fall 2026) Problem Set # 5
{{Your name}} **Due: November 2nd**

Problem 1. Markov Properties.

Let $G = ([n], E)$ be a simple undirected graph, and let $X = (X_1, \ldots, X_n)$ have a joint distribution μ, not necessarily strictly positive, on finite state spaces. For $A \subseteq [n]$, write $X_A = (X_i)_{i \in A}$, and let $\partial_i = \{j \in [n] : \{i, j\} \in E\}$ denote the neighbors of vertex i. Let $U \perp V \mid W$ denote that U and V are conditionally independent given W.

Recall the following definitions from lecture:

- **Global Markov property.** For any pairwise disjoint sets $A, B, S \subseteq [n]$, if every path from a vertex in A to a vertex in B passes through S, then $X_A \perp X_B \mid X_S$.
- **Local Markov property.** For every $i \in [n]$, conditioned on its neighbors, X_i is independent of all remaining variables: $X_i \perp X_{[n]\setminus(\{i\}\cup\partial_i)} \mid X_{\partial_i}$.
- **Pairwise Markov property.** For every pair of distinct, nonadjacent vertices $i, j \in [n]$, $X_i \perp X_j \mid X_{[n]\setminus\{i,j\}}$.

1. Prove that the global Markov property implies the local Markov property.
2. Prove that the local Markov property implies the pairwise Markov property.

   Hint: you may first show that $U \perp (V, W) \mid Z \Longrightarrow U \perp V \mid (Z, W)$.

Problem 2. Higher-order Tensor Decomposition.

Let $u \odot v$ denote the Khatri–Rao product of two vectors: if $u \in \mathbb{R}^m$ and $v \in \mathbb{R}^n$, then

$$u \odot v = \mathrm{vec}(uv^\top) \in \mathbb{R}^{mn},$$

where vec flattens a matrix column by column.

The *Kruskal rank* (or k-rank) of a collection of vectors $u_1, u_2, \ldots, u_m \in \mathbb{R}^n$ is the largest $k \in \{0, \ldots, m\}$ such that every subset of k vectors is linearly independent.

In this problem, we explore properties of the Khatri–Rao product and use them to design algorithms for decomposing higher-order tensors.

(a) Let k_u and k_v be the k-ranks of $u_1, u_2, \ldots, u_m$ and $v_1, v_2, \ldots, v_m$, respectively. Assume all u_i and v_i are nonzero. Prove that the k-rank of

$$u_1 \odot v_1,\ u_2 \odot v_2,\ \ldots,\ u_m \odot v_m$$

is at least $\min(k_u + k_v - 1, m)$.

(b) Construct a family of examples where the k-rank of

$$u_1 \odot u_1,\ u_2 \odot u_2,\ \ldots,\ u_m \odot u_m$$

is exactly $2k_u - 1$, and not any larger. To make this nontrivial, your examples must satisfy $m > 2k_u - 1$.

(c) Given an $n \times n \times n \times n \times n$ fifth-order tensor

$$T = \sum_{i=1}^{r} a_i^{\otimes 5},$$

give an algorithm for finding its factors that works for $r = 2n - 1$, under appropriate conditions on the factors $a_1, a_2, \ldots, a_r \in \mathbb{R}^n$.

Hint: Reduce to the third-order case.